

Industry Problem Task – Complete Solution

INDUSTRY PROBLEM TASK – COMPLETE SOLUTION

TASK 1 – DATA PREPARATION & INDUCTIVE LEARNING

(a) Use supervised inductive learning because labelled examples are available. The target is Diabetes: 0 = non-diabetic and 1 = diabetic. Key features: 1. Glucose – strong metabolic indicator of diabetes risk. 2. BMI – captures obesity-related metabolic risk. 3. Age – diabetes prevalence generally changes with age. 4. Blood Pressure – useful associated cardiovascular/metabolic feature. 5. Family History – captures inherited risk.

(b) Class imbalance should be handled without destroying useful medical records. SMOTE is suitable because it creates synthetic minority-class training examples. It must be applied ONLY to the training set, never to the test set. Class weights can also be used. Undersampling is less attractive because it discards majority-class observations.

(c) Use a stratified 70:30 train/test split. 70% provides enough training examples while 30% provides an independent evaluation set. Stratification preserves class proportions. Standardize numerical features for Logistic Regression; Decision Trees do not require normalization. Binary FamilyHistory needs no scaling for a tree and can remain 0/1.

TASK 2 – DECISION TREE

(a) Train a Gini-based Decision Tree. Gini impurity is: $Gini(S) = 1 - \sum p_i^2$ The root is the feature producing the largest impurity reduction. With the supplied synthetic data, the exact root and split values are produced by `Python_Code.py`. The first three tree levels are also generated as an image.

(b) Evaluate Accuracy, Precision, Recall and F1. False Negatives are actual diabetic patients predicted as non-diabetic. In screening, false negatives are important because they may delay confirmatory testing and follow-up. To reduce them, use class weights/SMOTE, threshold tuning toward sensitivity, and cost-sensitive learning. This may increase false positives, so sensitivity, specificity, precision and calibration should all be assessed.

(c) Pre-pruning restricts depth or leaf size before the tree grows. Post-pruning grows the tree and removes weak branches using cost-complexity pruning. A simpler pruned model is generally preferable when performance is similar because it reduces overfitting and is easier to interpret. In a clinical setting, selection should consider recall, calibration, robustness and external validation rather than accuracy alone.

TASK 3 – STATISTICAL LEARNING

(a) Logistic Regression is a statistical learning baseline. It estimates diabetes probability using a weighted combination of features. Compare Accuracy and AUC-ROC with the Decision Tree using the generated output. Logistic Regression is preferable when relationships are approximately additive, stable probability estimates and transparent feature effects are desired.

(b) Decision Tree importance is based on impurity reduction. Logistic Regression importance can be compared using absolute standardized coefficients. Glucose and BMI may rank highly in both models, but rankings can differ because trees capture nonlinear splits/interactions while logistic regression models additive effects.

TASK 4 – REINFORCEMENT LEARNING

(a) MDP: States = Stable, AtRisk, HighRisk, Improving. Actions = Diet, Exercise, Medication, Monitor. Reward = positive for movement toward a healthier state, lower reward for monitoring, negative reward for unsuitable transitions. Transitions = how an action changes the abstract health state.

(b) Q-Learning: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$ Use $\alpha=0.5$, $\gamma=0.9$ and $\epsilon=0.15$. The Python program runs at least five episodes and prints Q-table updates. A reasonable convergence criterion is maximum absolute Q-value change below 0.01 for five consecutive updates.

Example updates: $Q(\text{HighRisk}, \text{Medication})$: 0 $\rightarrow$ 2.5 after reward 5 when future value is initially 0. $Q(\text{AtRisk}, \text{Diet})$: 0 $\rightarrow$ 1.5 after reward 3 with zero future value. $Q(\text{AtRisk}, \text{Exercise})$: 0 $\rightarrow$ 1.5 after reward 3 with zero future value. Subsequent iterations include the discounted future Q-value.

(c) The final policy chooses $\arg\max_a Q(s,a)$ for each state. In real healthcare, an RL agent should not autonomously prescribe treatment. Patient safety, bias, explainability, privacy, reward misspecification, distribution shift and accountability require clinician oversight, offline validation and hard safety constraints.

IMPORTANT: The dataset and MDP used by the project are educational simulations, not clinical tools.